Startup Ecosystem Analytics Platform

Department of Computer Science and Engineering

Team Members

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Project Vision – Platform Objective

Platform Objective

Create a technical intelligence platform mapping founders, startups, and investor relationships.

- **Graph Topology:** Models natural networks without joining relational tables.
- **Safe Capital Flow:** Enables programmatic funding commitments on wallets.

Primary Goals

- **Query Latency:** Replace slow recursive SQL JOIN operations.
- **Concurrency:** Eliminate double-allocation conflicts on investment rounds.
- **Throughput:** Deliver search matching recommendations in sub-15ms.

Project Vision – Portal Roles

Startups & Investors

- **Startups:** Manage profile attributes, request round commitments, and showcase verified milestones.
- **Investors:** Track preferred sectors, manage transaction wallets, and receive match leads.

Analysts & Admin Users

- **Analysts:** Oversee database health, approve profile registrations, and verify achievements.
- **System Admin:** Audit wallets, track system health, and clear stale cache entries.

Problem Statement – Relational Pitfalls

SQL Database Bottleneck

- Relational tables require heavy JOINs to map multi-level connections.
- Computing match scores across 3rd-degree networks causes database locks.
- Performance degrades exponentially as nodes increase.

Lack of Transaction Protection

- Concurrent investment requests to the same round cause oversubscription.
- Race conditions lead to wallet balances drifting out of sync.
- Manual interventions needed for database correction.

Problem Statement – Workflow Obstacles

Unverified Achievement Lifecycles

- Founders self-report milestone metrics without validation.
- Inability to trace, audit, or verify historical round data points.
- Leads to investor distrust and inaccurate leaderboard states.

API Latency & Blocking Threads

- Heavy matchmaking computations block the server main request loop.
- Slow analytical lookups cause UI components to freeze or timeout.
- Limits platform scalability under concurrent user loads.

Proposed Solution – Graph Database (Neo4j)

Native Graph Storage Schema

- Maps entities (Founders, Investors, Startups) as nodes.
- Models actions (invested, interested) as physical graph edges.
- Eliminates the mismatch between application models and storage.

Fast Structural Traversals

- Evaluates path matching queries in microseconds.
- Computes network similarity scores without SQL JOIN overheads.
- Leverages index-free adjacency for relationship walks.

Proposed Solution – Memory Cache (Redis)

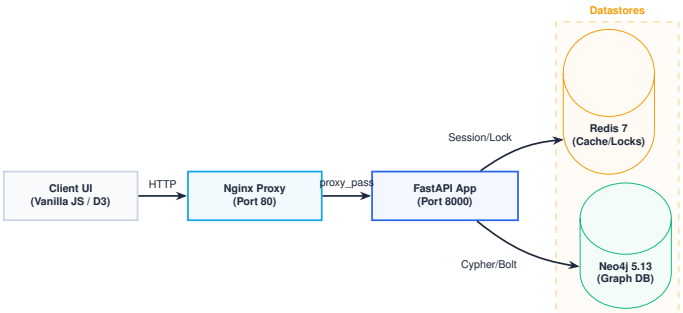
Caching & Result Set Buffering

- Caches heavy query results indexed by MD5 parameter hashes.
- Automatically invalidates profiles on data mutations.
- Serves read-heavy client requests in under 2ms.

Distributed State Synchronization

- Uses Redis lock leases to synchronize simultaneous transfer requests.
- Sorted Sets keep live ranking leaderboards updated.
- Tracks active sessions and coordinates distributed resources.

High-Level System Architecture



Network Isolation

Secure private network isolation hides Neo4j and Redis from direct public traffic.

System Architecture – Gateway Layer (Nginx)

Traffic Routing Configuration

- Exposes a single public IP mapping paths on Port 80.
- Reverse-proxies dynamic routes downstream to Python web-workers.
- Secures APIs by filtering unexpected request protocols.

Static Resource Caching

- Serves static asset files directly from host volumes.
- Decreases backend request load by bypassing Python runtime.
- Configured with gzip compression to minimize page load times.

System Architecture – Async Controller (FastAPI)

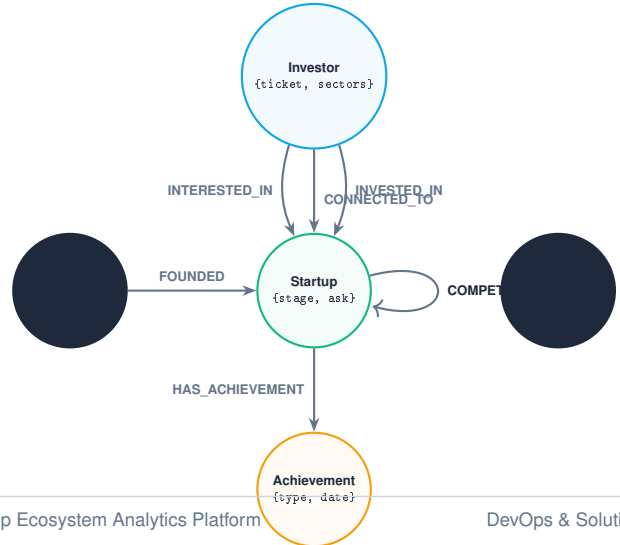
Asynchronous Route Operations

- Runs routes using Python's async-await concurrency framework.
- Implements dependency injection for connection management.
- Uses Pydantic to validate input JSON schemas strictly.

Connection Pool Management

- Connects to graph databases using persistent connection pools.
- Lifespan handlers bootstrap datastores and close drivers cleanly.
- Recycles dead database sessions automatically.

Database Schema – Graph Entity Model



Graph Properties

Modeling relationships as native entities allows the database to store property fields directly on edges.

Database Schema – Edge Relationship Properties

Attributes on Graph Edges

- Edges store crucial fields directly (e.g. `amount` on `INVESTED_IN`).
- Eliminates auxiliary lookup tables for relational data.
- Keeps historical transaction metadata close to nodes.

Structural Network Proximity

- Captures competitive relationships via `COMPETES_WITH` loops.
- Enables tracing of co-investment networks through shared nodes.
- Allows sector path traversals without index lookups.

Database Schema – Entity Constraints

Database Integrity Constraints

- Uniqueness rules configured on Startup IDs and Investor IDs.
- Prevents identity duplicates at the database server level.
- Restricts orphan relationships from corrupting the graph.

Automatic Execution Script

- Constraints created programmatically at API startup.
- Validates database schema structures during container bootstrap.
- Logs error states if constraints fail validation.

Database Schema – Query Indexing

Fast Property Lookup Indexes

- Secondary indices created on startup sector and funding stages.
- Speeds up match filters by avoiding full graph scans.
- Indexes investor preferred locations dynamically.

Performance Improvements

- Node property indexing reduces lookup times to sub-millisecond rates.
- Allows the engine to scale efficiently to thousands of nodes.
- Dynamic query planning benefits from updated indexing stats.

Core Engine – Hybrid Matchmaking Algorithm

Sector & Capital Alignment

- **Sector Match (40 pts):** Sector lists overlapping investor interests.
- **Ticket Overlap (30 pts):** Funding ask falls within investor ticket limits.
- **Stage Focus (20 pts):** Capital rounds align with investor focus stage.

Network & Velocity Enhancements

- **Network Bump (10 pts):** Co-investment paths are within two hops.
- **Milestone Velocity (10 pts):** Quick achievement velocity (≥ 3 milestones in 90 days).
- **Total Score:** Max score is 110 points.

Matchmaking – Cypher Query Structure

Match Scoring Execution

```
MATCH (i:Investor), (s:Startup {id: $startup_id})
WITH i, s,
  CASE WHEN s.sector IN i.preferred_sectors THEN 40
       ELSE 0 END AS sector_score,
  CASE WHEN i.ticket_min <= s.funding_ask
        AND s.funding_ask <= i.ticket_max THEN 30
       ELSE 0 END AS ticket_score,
  CASE WHEN s.stage IN i.stage_focus THEN 20
       ELSE 0 END AS stage_score
OPTIONAL MATCH path = (i)-[:INVESTED_IN*1..2]->(other:Startup)
WHERE other.sector = s.sector
WITH i, s, sector_score, ticket_score, stage_score,
  CASE WHEN count(path) > 0 THEN 10 ELSE 0 END AS network_score
OPTIONAL MATCH (s)-[:HAS_ACHIEVEMENT]->(a:Achievement)
WHERE a.date >= date() - duration({days: 90})
WITH i, sector_score, ticket_score, stage_score,
     network_score,
  CASE WHEN count(a) >= 3 THEN 10 ELSE 0 END AS ach_score
RETURN i.id, (sector_score + ticket_score + stage_score +
              network_score + ach_score) AS total_score
  ORDER BY total_score DESC LIMIT 10
```

Execution Profile

- Logic executes inside the Neo4j query planner.
- Minimizes runtime CPU load on Python web workers.
- Traverses up to 2-hop investor relations in microseconds.

Matchmaking – Cypher Execution Details

Cypher Case Optimization

- Computes logical scores entirely in the database engine memory.
- Minimizes CPU usage on the FastAPI web-app worker nodes.
- Uses pre-compiled query templates for speed.

Dynamic Path Traversals

- Traverses co-invested paths (`INVESTED_IN*1..2`) recursively.
- Caps returned matched sets to minimize serialization overhead.
- Avoids memory pagination issues with database LIMIT clauses.

Concurrency Control – Double-Spend Vulnerabilities

The Oversubscription Threat

- High-frequency investor transfers can exceed a startup's round limit.
- Simultaneous writes read stale balances before committing values.
- Leads to over-allocation of funding equity.

Relational Isolation Gaps

- Graph transaction scopes alone do not lock transient state across threads.
- Distributed locks are required to serialize transfer actions.
- Avoids deadlocks by applying timeouts to connection queues.

Concurrency Control – Two-Tier Protection

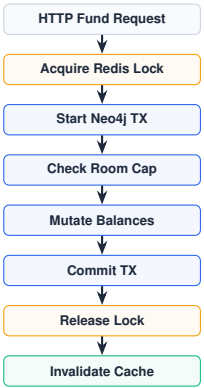
Distributed Locking (Redis)

- Backend acquires a lock on the startup ID before checking balances.
- Sets a 10s auto-expiry lease to prevent deadlocks on worker failures.
- Uses redlock principles to handle distributed clusters.

ACID Graph Transactions

- Transfers mutate balances and update graphs inside one session.
- Rejects transaction changes if the startup round limit is breached.
- Rollback triggers automatically on database constraint errors.

Transaction Lifecycle – Wallet Flow Sequence



Operational Flow

Synchronizes parallel transactions to ensure round data limits are preserved.

High-Performance Cache – Result Buffering

Query Hashing (MD5)

- Generates unique keys by hashing query parameters.
- Caches responses under `feed:filtered:{hash}` for 120 seconds.
- Limits CPU-intensive recommendations on repeated searches.

Active Invalidation Flow

- Any profile update triggers dynamic cache clearing.
- Keeps search data consistent without long wait cycles.
- Evicts dependent queries via atomic Redis keys deletes.

High-Performance Cache – Analytics & Leaderboards

Investor Rankings (ZSET)

- Redis Sorted Sets track scores on `leaderboard:investors`.
- Increments scores instantly on successful transaction commitments.
- Returns top ranked lists to dashboards with zero query latency.

Profile Views Tracking

- Increments click counters using Hash fields (`views:startup:{id}`).
- Periodic background tasks sync clicks back to the database.
- Mitigates database write loads from high-traffic profiles.

Security Architecture – Token Verification

Cryptographic JWT Engine

- Signatures validated using HMAC-SHA256 and custom secrets.
- Password structures secured with salted bcrypt hashing hashes.
- Enforces strict lifetime expiration validation on token payloads.

Token Revocation Lists

- Caches blacklisted tokens in Redis memory during logout actions.
- Checks incoming IDs against the blacklist during validation.
- Automatically evicts tokens using Redis TTL settings.

Security Architecture – Access Controls (RBAC)

Decorator-Based Authorization

- Restricts specific routes using custom decorators.
- Rejects request handling if user role claims are invalid.
- Validates scope hierarchies dynamically per request thread.

NoSQL Injection Protections

- Parametrizes all Cypher queries at connection execution.
- Rejects unescaped input characters during JSON parsing.
- Limits graph traversal query complexity at driver configuration.

DevOps Automation – Continuous Integration

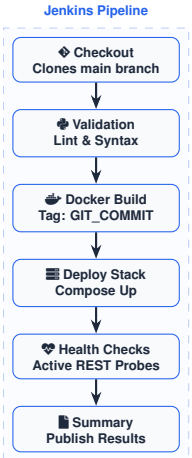
Lint & Syntax Verifications

- Automatically checks code syntax and formatting rules on PRs.
- Validates module package dependency structures.
- Uses auto-formatters to preserve consistent styling rules.

Pytest Verification Automation

- Runs the test suite in isolation using GitHub Actions.
- Validates backend routing and database driver connectivity.
- Rejects changes failing unit verification checks.

DevOps Automation – CD Deployment Pipeline



Continuous Delivery

Pipelines execute automated canary checks during deployments. Rollbacks run instantly if REST health signals timeout.

DevOps Automation – Container Orchestration

Container Isolation Strategy

- **Frontend Container:** Lightweight Alpine Nginx serves static resources directly.
- **Backend Container:** Python-slim execution environment hosting FastAPI.
- **Neo4j Container:** Persistent data folders mapped to host volumes.

Network Isolation Policies

- Locks backend containers inside private bridge networks.
- Only Nginx exposes ports to the external public network interfaces.
- Redis and Neo4j block direct external ingress ports.

System Verification – Pytest Infrastructure

Mock Testing Configurations

- Simulates graph drivers to test route logics in memory.
- Restricts external queries to ensure rapid execution speeds.
- Mocks JWT payloads to bypass login stages during endpoint checks.

Boundary Testing Cases

- Validates transaction locks under simulated parallel conditions.
- Asserts that unauthorized role access yields 403 HTTP codes.
- Tests limit compliance when wallet values edge close to zero.

System Reliability & Health Diagnostics

Active Diagnostics Probes

- Exposes a unified health route verifying external systems.
- Returns service connectivity flags and database latency.
- Enables standard container health checks dynamically.

Error Handling & Retries

- Middleware records request processing times and errors.
- Database drivers execute exponential backoff retry loops on fail.
- Re-establishes broken network connections without application restarts.

Future Roadmap & Project Summary

Technical Growth Roadmap

- Package container configurations as Kubernetes Helm charts.
- Implement central nodes search using Graph Data Science.
- Geo-replicate Redis caches to minimize global read latencies.

Architecture Conclusion

- **FastAPI Async Core:** High-concurrency operations.
- **Neo4j Graphs:** Relationships modeled natively inside graph stores.
- **Redis Buffers:** Sub-15ms search matching suggestions.